TECHNICAL NUGGET

3. Complexity Control: Lasso and Cross-Validation

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R. Tibshirani (1996). Regression Shrinkage and Selection via the Lasso. J. Royal Statistical Society, Series B, 58(1), 267–288.
J. Friedman, T. Hastie & R. Tibshirani (2010). Regularization Paths for Generalized Linear Models via Coordinate Descent. J. Statistical Software, 33(1), 1–22.

Is the Model Too Complex? Add a Penalty

$$\min_{\beta} \quad \tfrac{1}{2} \|y - X\beta\|^2 + \eta \sum_{j=1}^{d} |\beta_j|^p$$

p = 2 · ridge

$$\hat{\beta} = (X^{\mathrm{T}} X + 2\eta I)^{-1} X^{\mathrm{T}} y$$

Closed form: the normal equation with one extra term. Shrinks every coefficient, never to zero.

p = 1 · lasso (what we present)

$|\beta_j|$ has no derivative at 0, so there is no closed form. Needs an algorithm.

Sets coefficients exactly to zero: it selects features.

Start

The OLS sum of squares and its normal equation (Jana, section 2)

End

An update for one coefficient, repeated until nothing moves

Tool

Minimize over one coordinate at a time: a 1-D problem we can solve by hand

Derivation (1/2): One Coefficient at a Time

1. Standardize X, center y

The intercept drops out, and one η means the same for every feature.

2. Expand the sum of squares

$$\tfrac{1}{2}\|y - X\beta\|^2 = \tfrac{1}{2}\,\beta^{\mathrm{T}} G \beta \;-\; c^{\mathrm{T}}\beta \;+\; \tfrac{1}{2}\,y^{\mathrm{T}} y$$

$$G = X^{\mathrm{T}} X, \qquad c = X^{\mathrm{T}} y$$

3. Freeze every coefficient except β_j

$$f(\beta_j) = \tfrac{1}{2}\,G_{jj}\,\beta_j^2 \;-\; \rho_j\,\beta_j \;+\; \eta\,|\beta_j| \;+\; \text{const}$$

where

$$\rho_j = c_j - \sum_{k \neq j} G_{jk}\,\beta_k$$

ρ_j: how strongly column j lines up with what the other columns leave unexplained.

Derivation (2/2): The Soft-Threshold

4. Minimize f: convex, with a kink at 0

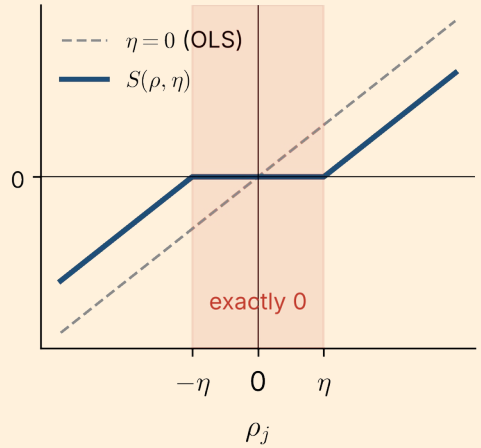
$\beta_j > 0$ $f' = G_{jj}\beta_j - \rho_j + \eta = 0 \;\Rightarrow\; \beta_j = \frac{\rho_j - \eta}{G_{jj}}$

valid only if $\rho_j > \eta$

$\beta_j < 0$ $f' = G_{jj}\beta_j - \rho_j - \eta = 0 \;\Rightarrow\; \beta_j = \frac{\rho_j + \eta}{G_{jj}}$

valid only if $\rho_j < -\eta$

$|\rho_j| \le \eta$ $\beta_j = 0$ neither case holds: the minimum sits on the kink



$$\beta_j \leftarrow \frac{S(\rho_j, \eta)}{G_{jj}}, \qquad S(\rho, \eta) = \text{sign}(\rho)\max(|\rho| - \eta,\ 0)$$

$\eta = 0$: $\beta_j \leftarrow \rho_j / G_{jj}$, which is Gauss–Seidel on the normal equation $G\beta = c$. No penalty returns OLS.

Pseudocode 1: Lasso by Coordinate Descent

LASSO(X, y, η) — *columns of X standardized, y centered*

$G \leftarrow X^T X; \quad c \leftarrow X^T y$ — *same two quantities as the normal equation*

$\beta \leftarrow 0$ — *start with every feature switched off*

repeat

  for $j = 1, \ldots, d$ — *d = 24 slopes*

    $\rho_j \leftarrow c_j - \Sigma_{k \neq j} G_{jk} \beta_k$ — *what column j still explains*

    $\beta_j \leftarrow S(\rho_j, \eta) / G_{jj}$ — *the 1-D minimizer from the derivation*

until max change in $\beta < 10^{-10}$ — *107 sweeps at $\eta = 0.05\, \eta_{max}$*

return β

$S(\rho, \eta) = \text{sign}(\rho) \cdot \max(|\rho| - \eta, 0)$ — *soft-threshold: zero when $|\rho| \leq \eta$*

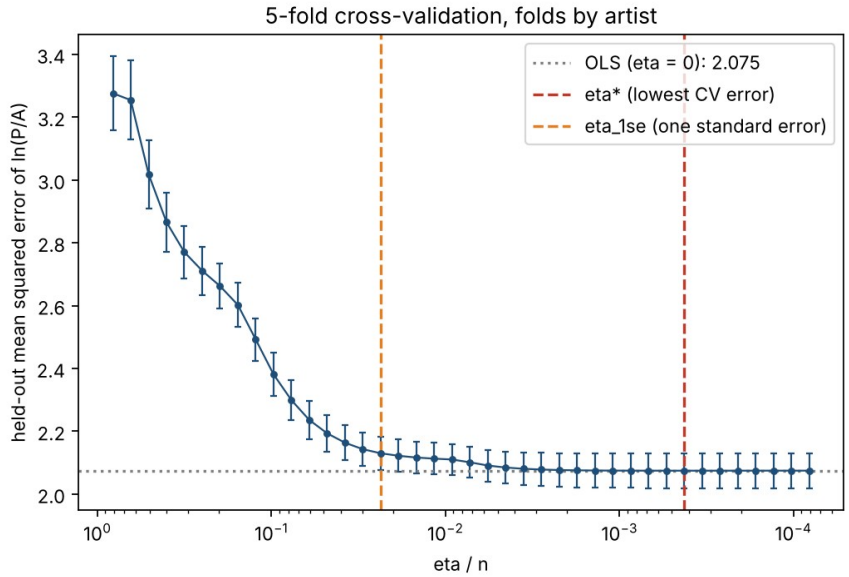
Checked in the notebook: $\eta = 0$ reproduces OLS, and $\eta > 0$ agrees with scikit-learn's lasso to 8 decimal places.

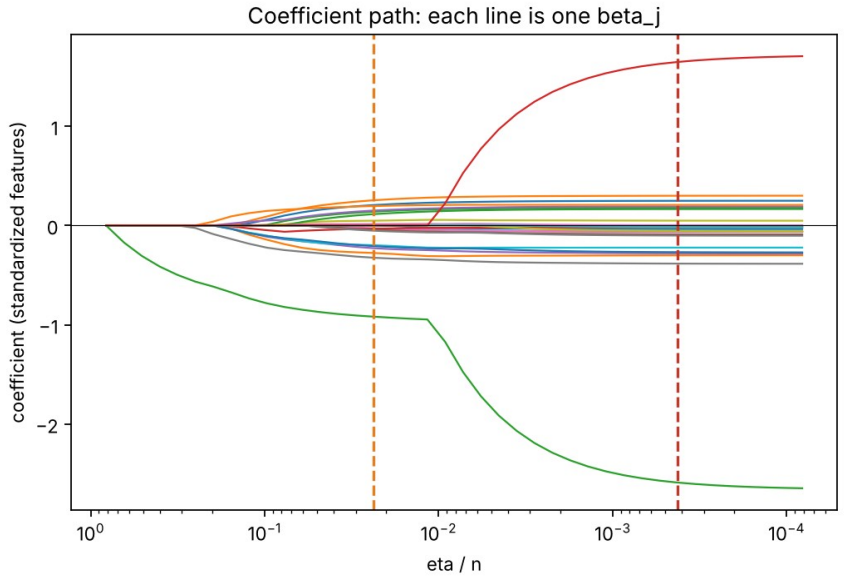
Pseudocode 2: Choosing η on Held-Out Data

CHOOSE-ETA(X, y, grid, K)	*K = 5; grid: 40 values from η_{max} down, plus 0*
assign every artist to one of K folds	*an artist's lots stay together*
for k = 1, …, K	
standardize using the other K − 1 folds	*held-out fold never touches the fit*
for η in grid, largest first	*$\eta_{max} = \max_j \lvert c_j \rvert$: every $\beta_j = 0$*
β ← **LASSO**(other folds, η)	*Pseudocode 1, warm-started from the last β*
err[η, k] ← mean squared error on fold k	
cv(η) ← mean over k of err[η, k]	
η* ← argmin cv(η)	*best predictive power*
return η*, **LASSO**(all data, η*)	*then compare β(η*) with β(0)*

Folds are by artist, not by lot: lots by one artist share fame and exposure, so splitting them would leak.

Demo: Sweeping η





Left: held-out error falls as the penalty is relaxed, then goes flat. The minimum is at η*/n = 0.0004.
Right: each line is one β_j. Features switch on one at a time as η shrinks.

Result: β(η*) Against β(0)

Coefficient	OLS, η = 0	Lasso, η*	Lasso, η1se	OLS p
Exposure: established	0.837	0.835	0.708	<0.001
Medium: drawing	−0.727	−0.722	−0.590	<0.001
Fame	0.133	0.133	0.126	<0.001
ln A	−2.078	−2.022	−0.717	<0.001
(ln A)²	0.085	0.081	**0**	<0.001
Support: panel	−0.111	−0.110	**0**	0.036
Created 2000+	−0.148	−0.138	**0**	0.050
Created pre-1800	−0.138	−0.123	**0**	0.251
Kept (of 24)	24	24	20	
R² on 2026 sales	0.380	0.381	0.372	

0.0001

gain in held-out error at η*, against a standard error of 0.056

−0.008

R² on 2026 sales from dropping 4 coefficients at η1se

Held-out data picks a penalty so small that β(η*) is essentially OLS.

What Complexity Control Tells Us

The model is not overfit

With about 900 sales per coefficient, held-out error is flat from η^* down to $\eta = 0$. The hedonic model's 24 slopes are supported by the data.

Limitation: collinear size terms

ln A and $(\ln A)^2$ move together, so lasso keeps one and drops the other. That is about the pair, not evidence that the size curve is a straight line.

Limitation: shrunken coefficients

Lasso coefficients are biased toward zero. Read them for which features survive; quote Jana's OLS table for effect sizes.

Next: if the linear model is not too complex, is it too simple? Section 5 tests that with a random forest.